What Do Benign-Trained Session Language Models Actually Detect?
A Mechanistic Audit with Corrected Evaluation Labels

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Abstract

Language models adapted to benign activity logs can rank insider-threat days highly, but a ranking score does not reveal what such a detector has learned: unusual behavior, profile novelty, and exposure during adaptation can all raise it. In this work, we audit Qwen3-8B adapters trained on non-insider employee logs from the CERT r4.2 and r6.2 insider-threat benchmarks, separating score construction, evaluation population, input sensitivity, and internal intervention effects. A calendar-label audit first corrects the evaluated populations to 1,362 positive days from 70 insiders (r4.2) and 73 days from 5 insiders (r6.2); the adapters' training membership is unaffected. Against negatives from the general user population, fold day ROC-AUC is 0.960 on r4.2 and 0.956 on r6.2; against benign users unseen during adaptation, it falls to 0.538 and 0.583. Because the score averages over every target token, profile fields included, we decompose it with the model and input held fixed: averaging only behavior targets raises r4.2 fold day ROC-AUC from 0.540 to 0.820, a score mitigation rather than evidence that the model ignores profiles. Inside the model, sparse-autoencoder edits of adapter-induced representation changes show predictive dependence without attack specificity: complete r4.2 ablation raises behavior loss but has a negative paired incident-window contrast, and fresh r6.2 behavioral-feature removal is not distinguished from norm-matched controls. We report each verified result with its scope and identify the repairs required before earlier causal claims can be reinstated.

1 Introduction

High ranking quality does not identify which information an anomaly detector uses. In employee activity logs, unusual behavior, profile novelty, and training exposure can all influence a score. We ask how these influences can be separated using frozen-model score decompositions, input manipulations, and internal interventions. This recovery edition distinguishes established measurements from claims still awaiting corrected experiments. Its contribution is an explicit audit and evidence record, not a detector leaderboard or a new mathematical theorem.

2 Data, labels, and populations

We study serialized employee activity from CERT r4.2 and r6.2 (Lindauer, 2020; Glasser & Lindauer, 2013; Le et al., 2020). CERT is a synthetic insider-threat benchmark. A record is one user's observed day, assembled from extracted sessions. The `DAY` line contains week and organizational fields, `PSY` contains simulated personality scores, and `SES` lines contain session activity counts. `SESCOUNT` records the number of sessions. These are structured measurements written as text.

Our primary target is membership in the official master's incident window: a day is positive if its recorded calendar interval intersects the user's half-open incident interval $[\text{start}, \text{end})$. Evaluation is restricted to days with extracted sessions. This labels an *incident-window day*, not every action on that day. An insider's other days are called *window-negative*; they are not verified benign activity.

An audit found an incorrect calendar join in the previous evaluation labels. The corrected day-index origin is December 28, 2009. Independent enumeration reproduced every corrected label, and the original texts, ordering, and splits were preserved. The resulting observed populations are:

Dataset	Observed days	Positive days	Insiders	Changed labels
r4.2	330,295	1,362	70	407
r6.2	1,393,297	73	5	23

The r4.2 changes comprise 177 positive-to-negative and 230 negative-to-positive labels; r6.2 has 10 and 13, respectively. The master windows include 530 r4.2 and 26 r6.2 additional days with no extracted session, which are outside this observation domain. Named malicious-event days define a different target; they are retained as sensitivity labels and are not silently substituted for the master-window target. The r6.2 master/detail discrepancies and one repaired malformed master date are documented in the label audit.

Training exposure. The main LM adapters remain usable: their actual training and validation user lists exclude all 70 r4.2 and all 5 r6.2 official insiders. This was checked independently of the faulty day-label join. The main training/validation counts are 851/79 non-insider users for r4.2 and 3,590/405 for r6.2. The unseen-user evaluation pools contain all official insiders plus those validation users: 42,468 days from 149 users and 142,072 days from 410 users. "Unseen" means absent from adaptation; these repeatedly analyzed validation users are not a newly untouched test set. The factorial and smaller-model adapters are separate training runs, with their own provenance qualifications.

Why labels affect more than a metric. Correcting labels can change a feature's rank, the sample used to fit a sparse dictionary, or which user is eligible as a normal donor. In those cases the historical intervention remains a measurement of its original inputs, but it is not the experiment the corrected procedure would have selected. Model training is reusable only when its actual inputs and fitting membership are unchanged. We apply this distinction throughout.

3 What the score measures

For tokens $x_1, \ldots, x_n$, the adapted LM assigns the anomaly score

$$s(x) = -\frac{1}{n-1} \sum_{t=2}^{n} \log p_{\text{ada}}(x_t \mid x_{<t}). \tag{1}$$

The first token is context, padding is excluded, and losses are assigned to the *predicted target token*. A larger score means the observed day was less predictable under the model of the adaptation corpus.

We separate two questions. In the general-population comparison, the negative user pool includes users seen during adaptation. In the unseen-user comparison, negative users are restricted to validation users absent from adaptation. Each fold contains one official insider and a fixed sampled negative-user cohort; metrics are averaged equally over insiders. The general comparison uses 800 non-insider users per fold; the unseen comparison uses all 79 or 405 validation users. An insider's window-negative days are negatives for day metrics. User scores are the maximum over each user's full scored history. The model stays fixed across these evaluation folds.

ROC-AUC measures the probability that a positive example outranks a negative one, with half credit for ties. Average precision (AP) summarizes precision across recall levels. It is not precision at a chosen alert threshold. A fold average and a pooled metric have different weighting and prevalence: neither can be substituted for the other. Unless explicitly labeled pooled, the tables below report equal-insider fold means. Bootstrap intervals resample users and are descriptive conditional on fixed models, settings, and negative cohorts; five r6.2 insiders provide particularly limited uncertainty information.

Changing scored targets. For a target-token class c, let $L_c(x)$ be its summed loss and $N_c(x)$ its number of scored targets. Then

$$s(x) = \frac{\sum_c L_c(x)}{\sum_c N_c(x)}, \qquad s_B(x) = \frac{\sum_{c \in B} L_c(x)}{\sum_{c \in B} N_c(x)}. \tag{2}$$

Our behavior view includes `SESCOUNT`, `SES`, and any residual non-profile target classes; the profile view includes `DAY` and `PSY`. In this score-only comparison the full original input remains visible to the model. A behavior target can therefore still be predicted from profile information. Removing profile targets from an average, removing profile *input*, and retraining without profile content are three different interventions.

4 Opening the model

The interpretability method remains part of the work. Its purpose is to connect an internal change to a measured output effect, while checking what the edit actually changed. Its historical empirical claims require the repairs listed later; the procedure itself does not depend on a favorable result.

4.1 Adapt the model, then hold it fixed

We adapt Qwen3-8B on non-insider user-days using QLoRA (Qwen Team, 2025; Hu et al., 2022; Dettmers et al., 2023). The base weights are frozen and quantized; small low-rank matrices supply the learned updates. For a projection matrix, the adapted weight is $W' = W + (\alpha/r)BA$. This makes it practical to compare the same model with adapters enabled and disabled. The model is frozen while we audit it. QLoRA supplies an accessible adaptation difference; it does not establish that an LM is the best detector for these data.

4.2 Isolate the adaptation difference

At token position t and layer ℓ, compare the adapted and base states:

$$\delta_t = h_t^{(\ell),\text{ada}} - h_t^{(\ell),\text{base}}. \tag{3}$$

This is the representation difference induced by adaptation on that input. It is not automatically a unique causal decomposition of the network.

4.3 Find a sparse approximation

Standardize each coordinate of δ_t and fit a sparse autoencoder (SAE):

$$q_t = (\delta_t - \mu) \oslash \sigma, \qquad z_t = \text{TopK}(\text{ReLU}(Eq_t + b_e)), \qquad \widehat{q}_t = Dz_t + b_d. \tag{4}$$

Only a few coefficients of z_t are retained. The dictionary columns of D are candidate directions of change, learned to reconstruct the observed deltas (Cunningham et al., 2023; Gao et al., 2024). Sparse approximation is not a proof that these directions are unique concepts. Any cross-dictionary comparison must account for normalization: physical directions are $\sigma \odot D_{:,f}$.

4.4 Select candidates, edit, and measure

An activation difference between incident-window and reference examples can rank candidate features. Selection is a hypothesis-generating step: labels, reference-user eligibility, and the dictionary sample all affect the answer. To test a selected set, edit its coefficients, continue the forward pass, and measure the change in a declared output such as behavior-target NLL. Compare against active features and random directions with the same edited support and matched realized perturbation size. Keep a same-configuration unedited forward pass as the reference. We measure both the requested edit and the change that actually survives low-precision rounding.

A change in output establishes sensitivity to that specified edit. It does not by itself prove that the features encode maliciousness, that the edit improves anomaly ranking, or that the selected set is strictly necessary. Best-of-search edits also differ from an intervention policy chosen before seeing its outcome.

4.5 Locate activation without overinterpreting it

For a fixed set of examples, activation mass at input class c is

$$m_f(c) = \frac{\sum_{x,t:\,\mathrm{class}(t)=c} z_{t,f}(x)}{\sum_{x,t} z_{t,f}(x)}. \tag{5}$$

This describes where a feature activates. A session-position hidden state can use an earlier profile. High profile-position mass in selected features is neither a complete account of the detector nor proof of a profile-only causal mechanism. Corrected attribution must use the corrected selected features and example population. Corrected r4.2 and fresh canonical r6.2 results are reported below; the r6.2 primary features have very little held-out activity, and the earlier cross-dataset profile-versus-behavior headline is not restored.

5 Technical specification of the mechanistic audit

5.1 Adaptation and representation

The main adapter configuration uses Qwen3-8B, NF4 four-bit base weights with double quantization, bfloat16 computation, and rank-16 LoRA updates with scaling 32 and dropout 0.05 on attention and MLP projections. The declared adaptation configuration uses maximum length 2,048, one epoch, learning rate 1.5×10^{-4}, cosine decay, and 3% warmup. Base and adapted passes use the same input and tokenizer. Training configurations and artifact hashes belong to each run; factorial adapters cannot be substituted for the main adapters merely because their architecture is the same. The recorded main r6.2 execution uses micro-batch 12, four ranks and one accumulation step: effective batch 48. The YAML default of eight is not the executed batch size for that adapter.

Let $\delta_t = h_t^{\mathrm{ada}} - h_t^{\mathrm{base}}$ at audited layer ℓ. With fitted location and scale vectors μ, σ, the SAE minimizes

$$\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \left\| (\delta_t - \mu) \oslash \sigma - \left(D\,\mathrm{TopK}(\mathrm{ReLU}(Eq_t + b_e)) + b_d\right)\right\|_2^2, \qquad q_t = (\delta_t - \mu)\oslash\sigma. \tag{6}$$

The implemented decoder is bias-free ($b_d = 0$); the fitted centering vector μ still enters reconstruction replacement. Both token sampling and optimization randomness must be recorded. The historical SAE seed argument did not seed torch initialization; earlier across-seed changes therefore mix sampling and optimization variability. The corrected fit records both. A dictionary fitted on evaluation examples is transductive even if feature ranking uses separate discovery users. It must not be described as a dictionary learned independently of every evaluation user.

5.2 Selection, attribution, and eligibility

A mean activation-gap screen proposes a feature set,

$$g_f = \mathbb{E}[z_f \mid y = 1, \text{discovery}] - \mathbb{E}[z_f \mid \text{declared reference}], \qquad S = \mathrm{TopFive}(g). \tag{7}$$

The reference population must be explicit: a window-negative insider day is not interchangeable with a non-insider day. Active controls avoid inert features but do not automatically match perturbation size. Token-weighted and equal-user summaries also answer different questions. Attribution mass describes the stated selected dictionary, token positions, and example set; changing any of these requires recomputation. Physical dictionary directions are $v_f = \mathrm{diag}(\sigma)D_{:,f}$, not standardized decoder columns alone.

5.3 Two distinct intervention implementations

Write $\widehat{\delta}_t = \mu + \sigma \odot (Dz_t + b_d)$ and choose a modified code z'_t. A residual-preserving additive edit is

$$h'_t = h_t^{\mathrm{ada}} + \sigma \odot D(z'_t - z_t). \tag{8}$$

A reconstruction-replacement edit instead sets

$$h'_t = h_t^{\text{base}} + \mu + \sigma \odot (Dz'_t + b_d). \tag{9}$$

Their difference includes the original SAE reconstruction residual. Even if two arms share replacement, downstream nonlinear responses need not cancel that residual's effects. Unchanged coefficients, unedited tokens, and the code bias must be handled consistently with the chosen implementation. Joint edits write the declared coefficients over a declared union of token positions; their interpretation is not a sum of proven single-feature effects.

For an output functional s_V on declared targets V, define $e_S(x) = s_V(x; I_S) - s_V(x; I_0)$, where I_0 is a live, same-configuration reference. A selected-versus-control contrast is $\mathbb{E}_u[\overline{e}_S(u) - \overline{e}_C(u)]$. An attack-specificity contrast compares this within-user difference across incident-window and matched window-negative receivers. Positive or negative signs must be interpreted using this output definition; an increased NLL is not automatically better anomaly ranking or successful mitigation.

Historical searched-donor repair used

$$\tau = (\overline{\Delta}^S_{\text{anomalous}} - \overline{\Delta}^S_{\text{noninsider}}) - (\overline{\Delta}^C_{\text{anomalous}} - \overline{\Delta}^C_{\text{noninsider}}), \tag{10}$$

with the best signed score change retained over searched donors and strengths. That defines a best-of-search estimand. A typical donor effect, a frozen intervention policy, and detector AP after intervention require separate measurements. The historical estimates are not reinstated in this edition.

5.4 Realized perturbations and numerical consistency

Let r_t be the requested float32 update and h_t the live low-precision state. The realized perturbation is

$$\widetilde{r}_t = \text{float32}\big(h_t + \text{cast}_{\text{BF16}}(r_t)\big) - \text{float32}(h_t), \tag{11}$$

where addition is also performed in BF16, matching the implemented hook. Matching $\|r_t\|$ does not guarantee matching $\|\widetilde{r}_t\|$. The corrected pilot checks edited support and realized norm per token at a fixed 5% relative tolerance, alongside no-op identity and loss-partition checks. Norm matching is one nuisance control; it does not make random directions semantically equivalent to selected features.

The historical scoring passes differed with batch shape and cross-entropy precision. Their per-example NLL variation, about 0.01 nats in the controlled checks, can exceed the intervention contrasts of interest. The corrected r4.2 reference uses batch size one and float32 cross entropy. Cached hidden states are not silently assumed equal to freshly computed states. Before reusing an effect, check adapter, tokenizer/text, dictionary and normalization, feature set, receiver, donor, token support, edit rule, score targets, and numerical configuration. A change in any of these can invalidate exact reuse.

5.5 Frozen cohorts and conditional mathematics

Fixed-input reanalysis retains the originally measured responses and changes only group labels. Post-hoc eligible-partner subsets have a different population from the original sample. Within-user contrasts use users with both corrected classes and do not recover the historical day pairing. All such estimates are exploratory. No coefficient-feasibility statement alone proves output mediation: a code-space target, its physically realizable hidden-state lift, and the model's behavior under that lift are distinct objects. The mathematical branch remains available for later work after these empirical dependencies are repaired.

6 Results supported after label correction

6.1 The comparison population changes the apparent detector quality

Dataset / negatives	Day ROC	Day AP	User ROC
r4.2 / general	0.960	0.011939	0.965
r4.2 / unseen	0.538	0.004978	0.589
r6.2 / general	0.956	0.000585	0.962
r6.2 / unseen	0.583	0.000123	0.649

The main-adapter day ROC changes from 0.960 to 0.538 on r4.2 and from 0.956 to 0.583 on r6.2 when negatives are restricted to unseen users. This is a population/exposure comparison, not a controlled intervention that identifies the sole cause of the gap. In r4.2, nine of the ten insiders newly included by the label repair are scenario-3 users, and their full scores rank incident-window days poorly. Their inclusion materially changes the main result; recovery cannot stop at the previously covered 60 users.

The historical Deep SVDD, recurrent-autoencoder, and Isolation Forest rows are not included as corrected comparisons. Their fitting populations need repair. The corrected LM measurements do not establish a detector leaderboard ordering.

6.2 Behavior-target scoring improves the complete r4.2 evaluation

r4.2 score	Day ROC	Day AP	User ROC
Full score	0.540	0.004859	0.589
Behavior targets	0.820	0.025238	0.825
Profile targets	0.482	0.001353	0.476

This separate, batch-size-one score decomposition covers all 42,468 unseen-pool days, all 1,362 positive days, and all 70 insiders. A 256-example bridge exactly reproduced the existing class-score pass before the missing users were scored. Class losses reconstruct the total score, and an independent check reproduced 840 reported fold metrics to floating-point precision.

With the model and input held fixed, changing the average from all targets to behavior targets raises mean fold day ROC from 0.540 to 0.820. The paired gain is 0.280, with a descriptive 95% user-bootstrap interval [0.204, 0.357]. AP changes from 0.004859 to 0.025238. This supports a *score mitigation* for this model and population. It does not show that profile information has stopped influencing behavior predictions.

Mean fold prevalence is 0.000714; pooled prevalence is 0.032071. The fold AP belongs with the former evaluation construction, not the pooled prevalence. Neither AP value is the percentage of alerts that would be correct at an operational threshold.

6.3 Training changes and score changes must stay separate

The next table concerns separate r6.2 factorial adapters, not the main adapters in the first table. Each input condition changes both adaptation and evaluation content, with one adapter seed per condition.

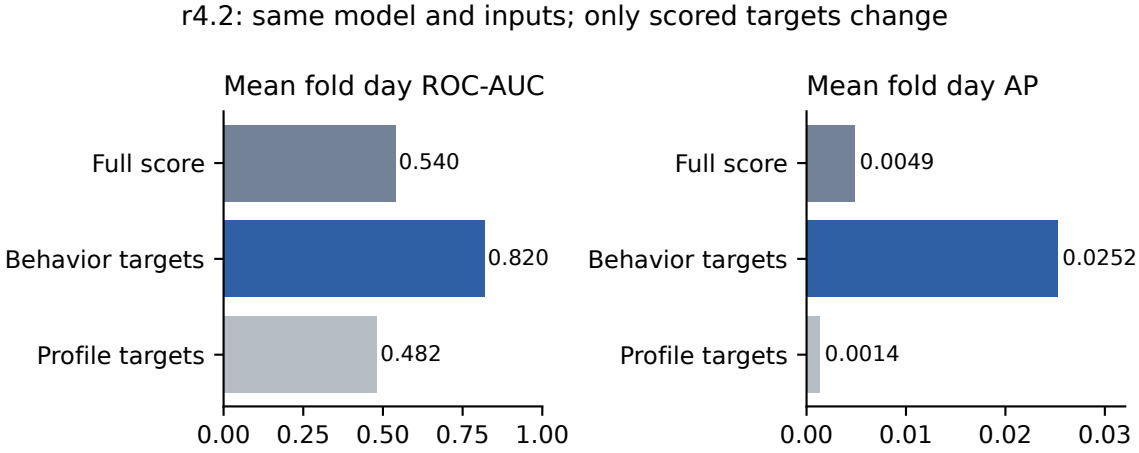


Figure 1: Complete r4.2 score decomposition. All three views retain the same input and frozen model. The figure reports fold means, not pooled metrics.

r6.2 adapter / input	Day ROC	User ROC
3B / Full	0.683	0.778
3B / No personality	0.799	0.901
3B / No profile	0.894	0.950
3B / Shuffled profile	0.432	0.550
8B / Full	0.412	0.472
8B / No personality	0.791	0.901
8B / No profile	0.899	0.902
8B / Shuffled profile	0.394	0.438

Removing all profile content improves ranking in both audited model families. The 3B change is substantial in point estimate; the earlier description of a uniformly tiny 3B effect is not supported. These are run-specific comparisons on five insiders, not a general model-capacity law. Their training-byte provenance is weaker than the main-adapter audit and remains explicitly qualified.

Scoring behavior targets while retaining the full input yields another comparison:

Separate r6.2 adapters	Full ROC	Behavior ROC	Behavior AP
3B full training	0.683	0.893	0.005480
3B target-mask training	0.350	0.899	0.003376
8B full training	0.412	0.897	0.001288

The 3B target-mask adapter omits profile targets from its training objective while retaining profile context. Relative to behavior scoring of its full-training counterpart, its day-ROC gain is about 0.006, with a descriptive interval crossing zero, while AP decreases. This does not establish an additional mitigation gain.

Input masking of frozen main adapters is yet another experiment. Removing personality input improves the cached day-ROC point estimates, but removing all profile input does not have the same effect. Those arms used different batch/hardware/cross-entropy configurations; the large personality-removal contrast exceeds the observed metric sensitivity, while small differences cannot be read at fine numerical precision. These findings are compatible with several mechanisms and do not identify one by themselves.

6.4 Temporal models provide a bounded comparison

The corrected cached numeric-model comparison uses a common population of 141,662 history-eligible unseen-user days:

r6.2 numeric model	Pooled day ROC	Pooled AP	User ROC
One observation	0.701	0.001391	0.776
Seven incl. target	0.708	0.002486	0.780
Forecast (past only)	0.726	0.001322	0.645

These are *pooled* metrics. The seven-observation reconstruction includes the target day; the forecast uses preceding observations only. Histories are previous available observations, not necessarily consecutive calendar days. All three checkpoint reloads passed the declared numerical tolerances, although the seven-observation profile score used 99.5% of its tolerance. That is a tolerance pass, not bitwise equality. History-shuffle and broader copying or mechanistic claims remain pending.

7 What the causal evidence currently establishes

7.1 The corrected pilot passes under a separate control policy

A corrected r4.2 pilot refit the sparse dictionary, reranked features using the corrected discovery labels, and scored 158 incident-window/window-negative pairs from 34 users under 15 conditions. The dictionary uses layer 26, 8,192 latents, and top-$k = 4$; its sample, fit, and feature selection are recorded. Live batch-size-one reference states and losses were recomputed. All 4,740 condition rows completed and the zero-edit controls reproduced the references.

One dictionary-direction control failed the predeclared 5% realized-norm matching tolerance at one window-negative token (6.42% mismatch). Consequently the original global gate remains failed. A separately versioned policy, declared after observing the original outcomes, replaces only dictionary-control draws that fail the same physical matching rule. A full replay completed all 4,740 rows and passed the validity and reproduction checks. Only the prescribed window-negative dictionary-control row changed; every unchanged measurement reproduced exactly. The three contrasts below therefore equal the original diagnostic estimates, while now being measured in a run that passes its own declared checks:

Policy-v2 contrast	Effect	95% interval
Selected minus isotropic	+0.005433	[+0.002277, +0.009191]
Selected minus active	+0.004373	[+0.001384, +0.007859]
Attack specificity	-0.003864	[-0.008807, +0.000146]

The first two contrasts show larger behavior-loss increases for selected edits than for their specified controls on incident-window receivers, but the specificity contrast does not establish a larger attack-specific response. These estimates do not reinstate the earlier causal headline. The replay is exploratory: the dictionary is transductive, the users were previously inspected, and the revised control policy was declared after the original results. Reproduction provides implementation validation, not independent confirmation. The intervals are unadjusted 95% user-bootstrap intervals; the selected-versus-active contrast compares two intervention policies rather than equal doses. The original failed run and its diagnostics remain preserved.

7.2 Corrected r4.2 features activate mainly on session tokens

Using the corrected dictionary and its fixed five discovery-ranked features, we recomputed activation location on all 1,362 incident-window days from 70 users, with 500 fixed val days from 79 official non-insiders as a reference. All 501,711 selected token positions passed coverage checks. The percentages below pool each feature's activation mass within each positive population; "active" means nonzero mass on at least one incident-window day.

Feature	Discovery SES (%)	Confirmation SES (%)	Active conf. users
3843	91.45	84.95	13/35
3230	100.00	100.00	32/35
3199	100.00	100.00	24/35
3975	100.00	100.00	30/35
1394	100.00	100.00	30/35

Four features place virtually all their positive-day mass on session tokens. Feature 3843 has a more mixed location and weak confirmation activation: its mean activation per token falls from 0.108165 on discovery days to 0.000244 on confirmation days. Thus a high session-location percentage does not establish a stable activation level across users. The dictionary remains transductive and these confirmation users are not a fresh holdout. Session states can use earlier profile context, so this result does not prove behavior-only information use or a causal circuit. The fresh r6.2 analysis below does not restore the earlier cross-dataset dissociation.

7.3 Fresh r6.2 features and their measured effects

We rebuilt the r6.2 analysis from a canonical extraction: one unpadded example per forward pass, with the frozen adapter and layer 18 unchanged. Five fresh sparse dictionaries use 16,384 features and top-$k = 8$. Each held-out user's labels are excluded from feature ranking, but their unlabelled inputs enter dictionary fitting; this is transductive, not fully inductive evaluation. The primary ranking selects the five largest discovery-positive versus non-insider activation gaps. The behavioral ranking uses session-token gaps among features with at most half their discovery-positive mass on profiles. Neither ranking uses the held-out user's labels. The fresh attribution results are:

User	Days	Disc. P (%)	Primary entries	Behav. entries	Held P (%)
ACM2278	6	99.88	2	14	0.00
CDE1846	45	43.97	18	211	0.26
CMP2946	20	99.92	0	31	–
MBG3183	1	99.99	0	12	–
PLJ1771	1	100.00	0	6	–

Both profile percentages refer to the primary set; discovery and held-out groups are separate. Entries count nonzero feature–token pairs, not distinct positions. Across 73 incident-window days and 23,082 positions, primary sets have only 20 entries and three zero-mass users, whose location fractions are undefined. The behavioral sets have 274 entries and nonzero mass in all five users. CDE1846's primary discovery set is mixed, so the primary ranking should not be called uniformly profile-specific. All 73 live positive delta grids and selected feature-class activity counts were checked against the accepted attribution artifacts. Historical cache results and their failed live compatibility check remain separate in Appendix A.2.

The intervention batch contains all 73 positive days and 28 unique same-user window-negative days, retaining 73 fixed pairs and repeated-negative weights. One hash-selected official-non-insider donor is fixed per receiver before scoring. Edits preserve the original residual state and remove selected decoded feature contributions, or patch their coefficients toward that donor's active mean. Zero activity remains in every relevant denominator. Loss is recomputed in the same canonical configuration, with parameters and observed targets fixed. At removal strength one, the equal-user behavior-target results are:

Family	Behavior-loss contrast	Effect	95% interval
Primary	Removal minus baseline	-0.000041	[-0.000114, +0.000000]
Primary	Removal minus matched direction	-0.000230	[-0.000679, +0.000000]
Primary	Removal minus active	-0.000515	[-0.001322, +0.000176]
Primary	Paired contrast: matched direction	-0.000201	[-0.000699, +0.000098]
Primary	Paired contrast: active	-0.000650	[-0.001401, +0.000110]
Primary	Fixed-donor patch minus baseline	-0.000036	[-0.000098, +0.000000]
Behavioral	Removal minus baseline	+0.000867	[+0.000056, +0.002304]
Behavioral	Removal minus matched direction	-0.000216	[-0.000939, +0.000556]
Behavioral	Removal minus active	+0.000528	[-0.000107, +0.001181]
Behavioral	Paired contrast: matched direction	+0.000737	[-0.001315, +0.003209]
Behavioral	Paired contrast: active	-0.000161	[-0.001824, +0.001250]
Behavioral	Fixed-donor patch minus baseline	+0.001705	[+0.000794, +0.002571]

Effects are differences in behavior-target loss, in nats per target, between the stated arms. A positive removal-minus-baseline effect means worse prediction after removal. "Paired contrast" subtracts the matched window-negative contrast from its incident-window counterpart, first averaging pairs within each user and then the five users equally. Dictionary directions match realized per-token support and norm within 5%; the natural active-feature control does not match dose. Fixed-donor patches are a separate policy; ablation controls do not establish patch specificity. Intervals use 10,000 five-user bootstrap draws and are descriptive, conditional on these models, dictionaries, rankings and pairs.

The primary features produce little held-out response: three users have no selected activity, and the full-dose removal mean is slightly negative. For the behavioral set, removal raises loss by +0.000867, but selected-minus-matched-direction is -0.000216, with interval [-0.000939, +0.000556]. The paired matched-direction interval also spans zero. Thus the observed removal response does not establish greater dependence on the selected features than on norm-matched directions, or attack-specific dependence. The largest behavioral removal response comes from PLJ1771, who contributes only one positive day. Fixed-donor behavioral patching raises, rather than lowers, loss on average (+0.001705); it does not demonstrate the intended score reduction. Neither family establishes improved detector ranking or a complete internal mechanism.

One primary quarter-dose dictionary control could not meet the unchanged matching rule within 32 candidates. A versioned reporting repair preserves that infeasible condition and continues the other measurements: 2,726 measured slots out of 2,727 planned, including the common zero condition. No receiver was removed, no missing effect was set to zero, and no tolerance was widened. All full-dose endpoints have complete coverage. Affected quarter-dose full-population estimates remain unavailable. The reporting revision followed the observed failure and is disclosed as such; numerical verification is not independent scientific confirmation. Per-user results, doses and the validation scope are given in Appendix A.1.

7.4 Complete r4.2 ablation supports predictive dependence, not attack specificity

The residual-preserving ablation run covers all 1,362 incident-window days and 380 unique matched window-negative days: 1,742 receivers under eight conditions, or 13,936 scored rows. Independent implementation checks passed. The frozen analysis averages days within users, then users equally, retaining zero edits and repeated negatives through the fixed pair ledger. The primary group contains 667 positive days from 35 previously inspected confirmation users. At fixed removal strength one, the behavior-target loss contrasts are:

Behavior-loss contrast	Effect	95% interval
Selected minus baseline	+0.004411	[+0.001812, +0.007283]
Selected minus matched direction	+0.005531	[+0.002886, +0.008625]
Selected minus active	+0.004275	[+0.001709, +0.007179]
Incident minus window-negative	-0.004599	[-0.009430, -0.000616]

The final row subtracts the matched window-negative selected-minus-active-control effect from the incident-window effect within each pair. Its negative estimate indicates a smaller relative response on incident-window

days under this policy, contrary to an attack-specific enhancement. Selected removal nevertheless raises behavior loss on the positive population relative to baseline and the stated controls. The dictionary-direction control matches realized per-token support and norm using the frozen feasibility policy; the natural active-feature control does not match dose. The intervals are unadjusted, descriptive 95% user-bootstrap intervals from 10,000 draws, conditional on the dictionary, controls and pairs. All 64 primary/secondary summaries and 2,240 user estimates were independently reproduced. The transductive dictionary and previously inspected users preclude an independent-confirmation claim. These loss changes do not establish detector mitigation, a complete circuit or the historical reconstruction-replacement result.

7.5 Saved responses retain a narrower meaning

The historical Pilot 2 profile-swap measurements were relabeled on CPU. Of its 414 receivers, 129 change class; 86 have an ineligible partner in at least one scored swap condition. The adapted model remains more sensitive than the base model to the specified profile swaps in the frozen cohort. This measures input sensitivity. Corrected class contrasts and eligible-partner subsets are post-hoc descriptions; they do not restore the original matching or identify harm, mediation, or a complete circuit.

The H6 direct-intervention pilot retains 1,320 scored rows for 120 receivers. Forty-eight receiver labels change, and 44 receivers have at least one official insider among their nominally benign profile partners. Its previously reported small pooled mean effects reproduce. Small means for two specified lifts do not establish equivalence of all feasible implementations or all users. The corrected class summaries are likewise limited to the historical inputs and selected dictionary.

7.6 Corrected input swaps: a separate sensitivity result

Fresh scoring now covers all 1,362 corrected positive r4.2 days and 73 r6.2 days. Each day receives one frozen, hash-selected donor from official-non-insider validation users: first choose a user, then one of that user's days. We replace static DAY fields and PSY while retaining the receiver's week and behavior. Original, profile-swapped and behavior-swapped inputs are scored with the adapter enabled and disabled on the same quantized backbone: 8,610 score rows. These synthetic inputs have no asserted attack ground truth.

The primary contrast is profile-swapped minus original negative log-likelihood on identical behavior targets, matched by relative character span and token ID. It excludes direct profile-target losses. No input was truncated and no behavior target was discarded. Average days within users, then users equally:

Dataset	Adapted	Base	Difference	95% interval
r4.2	+0.1535	-0.0008	+0.1543	[+0.1338, +0.1748]
r6.2	+0.1158	-0.0093	+0.1250	[+0.0864, +0.1637]

The last two columns give adapted minus base contrasts and descriptive 10,000-draw user-bootstrap intervals conditional on the fixed donors. All 70 r4.2 user means and all five r6.2 user means have positive adaptation contrasts. Five users provide limited information about population uncertainty.

Profile replacement changes behavior-token positions for 768 r4.2 and 45 r6.2 days. A post-hoc restriction to unchanged positions retains 594 days/63 users and 28 days/five users, respectively; its equal-user adaptation contrasts remain positive (+0.1670 and +0.1121). This checks a possible length explanation; it is not an independently confirmed endpoint.

The historical broad interpretation that profile swaps dominate behavior swaps is not restored. On full-sequence loss, equal-day profile/behavior changes are +0.1709/+0.1069 on r4.2 and +0.1686/+0.0732 on r6.2. Equal-user changes are +0.1211/+0.1628 and +0.1036/+0.1135, reversing that ordering. Moreover, behavior replacement changes the predicted targets, so these are not matched measures of mechanism importance. The supported result is profile-conditioned behavior prediction under this donor policy. It does not establish a harmful shortcut, attack specificity, SAE mediation, or robustness across donor policies, and it does not replace the required internal-intervention reruns.

7.7 Historical claims awaiting dependency repair

The old sparse-feature attribution percentages, repair/ablation estimates, cross-dataset transfer, seed robustness, benign-only dictionary replication, and smaller-model mechanistic replication are not presented as corrected findings. Their dictionaries, ranking, negative pools, receivers, or donors can depend on the faulty labels. For the repaired-3B study in particular, correcting missing positive examples is insufficient: its label-selected negative sample must also be redrawn. Historical artifacts are preserved for exact reuse checks, not discarded; reuse requires the same complete intervention definition.

8 Interpretation and remaining work

The strongest corrected result is presently about the relationship between *evaluation population, score construction, and input dependence*. A model can achieve a high ROC under one comparison population, perform much less well on unseen users, and improve when only behavior targets contribute to its score. Those observations motivate mechanistic investigation; they do not already settle it. Complete r4.2 ablation now shows a behavior-loss response to selected removal, while its paired incident-window contrast is negative rather than supporting attack specificity. Fresh r6.2 primary features have very little held-out activation. Behavioral-feature removal raises loss, but its contrast against norm-matched directions and its paired specificity contrast remain inconclusive. Fixed-donor behavioral patching raises loss instead of reducing it. These results constrain the mechanistic account and remain visible alongside the positive score-level findings.

The interpretability method remains useful: compare adaptation-induced states, fit a sparse approximation, select candidates on a declared discovery group, measure behavior-loss interventions against realized perturbation controls, and report attribution with its location-versus-meaning limitation. Its methodological value depends on executing these checks and exposing negative or inconclusive outcomes. Existing sparse coding, causal contrasts, and feasibility mathematics do not become new mathematics merely through application to insider threats. The mathematical branch is retained as conditional work, with no new theorem or novelty claim asserted here.

Corrected classical comparators. The classical probe has now been refitted after removing the ten r4.2 and one r6.2 official insiders from its nominally normal training pool. Isolation Forest (IF), one-class SVM (OCSVM), and PCA reconstruction scores each have separately fitted arms with ($+P$) and without ($-P$) profile features. The complete unseen-user evaluation includes 42,468 r4.2 days (1,362 positive) and 142,072 r6.2 days (73 positive). These are pooled day metrics, unlike the equal-insider fold means reported for the main LM:

Dataset / model	ROC +P	ROC −P	AP +P	AP −P
r4.2 / IF	0.752	0.759	0.068445	0.068818
r4.2 / OCSVM	0.641	0.663	0.116471	0.120560
r4.2 / PCA	0.789	0.801	0.150859	0.153274
r6.2 / IF	0.664	0.667	0.001014	0.001022
r6.2 / OCSVM	0.520	0.535	0.009153	0.006569
r6.2 / PCA	0.653	0.674	0.039796	0.039828

Removing profiles improves pooled day ROC in all six comparisons, but the AP and within-user results are mixed. For example, r6.2 OCSVM AP falls from 0.009153 to 0.006569. This is a comparison of separately fitted models, not an intervention on a frozen detector. The numeric day features also contain information absent from the LM serialization, so this table does not isolate the effect of architecture or establish QLoRA superiority. The accompanying column-permutation diagnostics are reported with their signed effects and scope in Appendix A.

Other comparison-model dependencies. For the local session autoencoders, an audit of actual saved bundles found contaminated training membership in 94/120 r4.2 bundles and 8/8 r6.2 bundles. All 120 r4.2 bundles have additional official insiders in their test groups. The 26 r4.2 bundles with clean training

membership remain candidates for limited reuse; they do not clear the old benchmark as a whole. The older sequence-baseline harness shares the faulty fold construction and saves aggregate metrics, without the checkpoints and per-example predictions needed for score-only recovery. Refits must save scaler membership, checkpoints, predictions, and corrected fold definitions before those baseline comparisons return.

Scope outside CERT. The CERT clock bug does not resolve separate TWOS or LANL issues. The early TWOS feature ranking had an independent defect; the corrected bounded held-out study remains inconclusive. LANL has an independent coupling between sampling and fold hashes. Neither is treated as a cleared replication here. Personality novelty associations on only five r6.2 insiders remain exploratory and are not used to explain the detector as a general law.

Next evidence needed. The next experiments must name the exact unsupported claim, corrected inputs, frozen endpoints, eligibility checks, output artifacts, and measured runtime before scaling. Priority goes to corrected causal selection and controls only where the retained paper claim needs them; comparator refits must first repair their training folds. Repeating every historical job is unnecessary, while relabeling every old output is insufficient. The 46-family recovery ledger records this distinction. This edition reports the verified subset and the remaining gaps; it is not a claim that the complete mechanistic program is finished or ready for submission.

9 Conclusion

Corrected labels preserve the main LM adapters but change several evaluated populations and invalidate some label-dependent experiment choices. Complete r4.2 behavior-target scoring improves ranking without changing the model or removing profile context. The interpretability question therefore remains: which internal pathways contribute to behavior predictions, and are their measured effects specific to incident-window activity? The validated exploratory pilot, complete r4.2 ablation and fresh r6.2 interventions do not establish that specificity. The r6.2 primary family also shows little transfer to held-out users. A defensible account keeps these limits visible while retaining the verified score-level findings.

A Recovery provenance and scope

The machine-readable source is `paper/data/recovery_registry.json`. It records all 46 claim families, source paths and SHA-256 digests, exact metric selectors, completed scopes, limitations, and remaining work. Shared tables, values, and the mitigation figure are generated from that registry for all three editions. Source audits and the old manuscript texts are preserved in the repair branch history and in the external recovery archive. Historical figures and tables are not part of the corrected build.

Label validation. The official answer archive checksum was checked. Separate code enumerated the master-window calendar target and compared all metadata rows; it did not import the production label builder. Every named malicious event falls inside the corresponding user's extracted session interval under the recorded calendar clock. The r6.2 master includes a malformed CDE1846 start field, repaired using that named user's earliest recorded event. Master/detail discrepancies for ACM2278, CMP2946, and PLJ1771 are recorded rather than resolved by mixing target definitions. Corrected evaluation JSONLs preserve all non-label values.

Score recovery. The cached registry preserves corrected fold means, pooled metrics, paired contrasts, and fixed-sample results separately. Duplicate score identities must agree before deduplication. Each score view must have finite values and valid target counts; class losses must reconstruct total loss. The complete r4.2 extension follows an exact 256-row numerical bridge. The same-configuration class pass and the older full-population pass differ slightly numerically, which explains why the r4.2 full-score value in the mitigation table is 0.540 rather than the first table's 0.538. They must not be spliced into one causal comparison.

New CPU recovery. Pilot 2 and H6 retain their saved losses and historical feature selection. An independent arithmetic implementation reproduced 1,680 summaries and interval endpoints within 6×10^{-17}; pooled H6 values also reproduce the earlier independent audit. This checks aggregation, not the validity of the historical sampling protocol. Donor manifests were checked against the official insider list, independently of observed positive-day labels. The autoencoder audit reads actual saved fold membership and hashes each bundle, rather than inferring contamination from the manuscript's description.

Corrected causal pilot. The new r4.2 dictionary was fitted for 20 epochs on two million sampled token deltas, with layer width 4,096 and 8,192 sparse features, top-$k = 4$. Feature ranking used corrected discovery labels. The dictionary remains transductive; previously inspected confirmation users are not fresh holdouts. The full 158-pair run took approximately 45.9 minutes on an RTX 3070. Its original global norm gate failed at one control token. Separate diagnostic analyses preserve that failure. Following an engineering slice, the full replay under `dictionary-control-bf16-rejection-v1` took 54 minutes 28 seconds, completed all 4,740 rows, and passed its declared checks. An independent CPU review reproduced 91 numerical summaries with maximum absolute difference 2.78×10^{-17}. These runtimes measure the pilot executions, not the cost of all historical claim recovery.

Input-swap audit. The corrected C10 study recomputes original and substituted inputs under both model states. A separate audit reconstructs the replacements from original JSONL texts, validates canonical labels and official-insider exclusions, retokenizes every input, and verifies exact-span target alignment. All 384 pilot score rows reproduce exactly at every token in the full runs. Separate arithmetic checks reconstruct loss-class sums and paired effects. Donors are globally sampled from validation users, not matched on role, department, chronology or behavior. The swap can create an unusual profile–behavior combination; sensitivity does not distinguish useful conditioning from an undesirable shortcut. One donor per day and receiver-user bootstraps leave donor-choice uncertainty unmeasured.

Classical-comparator recovery. Training membership excludes every official insider, independently of observed positive-day labels. The corrected fitting pools contain 287,827 r4.2 days and 1,251,225 r6.2 days. Scalers use training rows only. IF uses 300 trees with seed 42; OCSVM uses $\nu = 0.05$ and `gamma=scale` on the same fixed 6,000 training IDs in each arm; PCA uses ten components. Validation and evaluation users are absent from fitting. All their observed days, including insiders' window-negative days, enter the pooled metrics. Per-user results and the identities of maximum-score days are retained in the artifact bundle.

Each numeric column is independently permuted across all evaluation days under three frozen seeds, using identical permutations across the three with-profile models. The primary diagnostic is the signed, unrounded decrease in day ROC, averaged over the three draws for each column. The table sums those means within profile and behavior groups. Its last column is a secondary statistic: clip each column's mean decrease at zero, then divide the profile sum by the total positive sum (undefined when that total is zero).

Dataset / model	Profile sum	Behavior sum	Clipped share (%)
r4.2 / IF	-0.0066	+0.0576	1.23
r4.2 / OCSVM	+0.0113	+0.1047	10.25
r4.2 / PCA	+0.0057	+0.3404	3.39
r6.2 / IF	-0.0162	+0.0560	3.42
r6.2 / OCSVM	-0.0214	-0.0884	6.08
r6.2 / PCA	-0.0198	+0.2051	4.05

Negative sums mean that those marginal permutations improved ROC on average. In particular, positive clipped shares do not imply a positive signed profile effect. Correlated columns can share predictive information, and permutation can produce combinations outside the observed distribution. These sums are neither joint profile-block effects nor additive causal contributions. Three draws measure permutation variation, not population uncertainty; r6.2 still has only five independent incident users. Numeric feature groups and input swaps also change different objects, so their magnitudes are not directly comparable.

An independent reviewer replayed all 1,107,240 baseline checkpoint scores, hashed all 1,089 saved permutation arrays, reproduced every reported metric, and replayed 36 complete model/column/draw cases from checkpoints. This clears the declared C27 protocol, not the separate autoencoder or sequence-baseline families. Standardized arrays and raw permutation scores remain in the Aquaman evidence archive, with digests and reviewed summaries in the registry.

Pending work and reusable artifacts. Raw sessions, corrected texts, main-adapter weights, and fixed-input responses remain useful. A contaminated scaler or detector requires refitting; a changed SAE dictionary requires downstream re-encoding, feature ranking, attribution, and required interventions. The repaired-3B negative sample also needs correction. The optional five-user personality-novelty association is deferred rather than promoted as an explanation. TWOS, LANL, and conditional geometric results retain their separate ledgers. The companion claim map and rerun specifications explain what each retained or deferred result would require.

A.1 Fresh r6.2 intervention details and coverage

The fixed positive cohort has six ACM2278, 45 CDE1846, 20 CMP2946, one MBG3183 and one PLJ1771 day. Each positive is paired with a same-user window-negative day, preferring the same weekday, then minimum calendar distance and a fixed hash tie-break. Negative reuse is retained through the pair ledger. Donor selection first hashes into sorted validation non-insider users, then into that user's sorted days. The same donor is used for both feature rankings. No donor is chosen using an activation or loss response. Zero donor support gives a zero prototype under the declared rule.

For a selected feature set S, let $z_{t,S}$ be its receiver coefficients at input position t, and let $z^d_{j,S}$ be its donor coefficients at position j. Define the active position sets $A = \{t : \|z_{t,S}\|_1 > 0\}$ and $B = \{j : \|z^d_{j,S}\|_1 > 0\}$. The donor prototype is $p_S = |B|^{-1} \sum_{j \in B} z^d_{j,S}$ when B is nonempty and zero otherwise. With removal strength $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$, the requested edits are

$$r_t^{\text{remove}} = -\alpha\, \sigma \odot D_{:,S} z_{t,S}, \qquad r_t^{\text{patch}} = \mathbf{1}_{\{t \in A\}}\, \sigma \odot D_{:,S}(p_S - z_{t,S}). \tag{12}$$

Here $D_{:,S}$ contains the selected decoder columns, σ restores physical coordinate scales, and the indicator is one only on active receiver positions. Both edits are added to the original adapted state; patching uses strength one. The natural active control removes its own five ranked control features at the same nominal strength. The dictionary control uses other decoder directions, matched to the selected removal's realized support and norm at each position.

All runs use canonical unpadded batch-one NF4/BF16 SDPA evaluation with TF32 disabled. The SAE runs in FP32; shifts are restored to model coordinates before the recorded BF16 casting and addition. The independent replay checks saved coefficients, TopK rank/cardinality, CPU/GPU dense scores, exact support on every feature used by the edits, explicit-sum physical shifts, exact BF16 arithmetic, no-op loss identity, control RNG and 5% realized norms. Unrelated cutoff support exchanges are recorded under the declared finite-precision policy; no mismatch in an edited feature is accepted. This policy does not make the original global support check pass. The replay checks saved loss grids and their bindings; it does not independently rerun every edited LM prediction.

The full-dose user means below retain all five users, including inactive sets. "Removal" is selected minus baseline; "Vs direction" subtracts the norm-matched dictionary control; "Paired" is the incident-minus-window-negative difference of that contrast. "Patch" is the fixed-donor patch minus baseline. All columns score behavior targets, including session-count targets.

Family	User	Removal	Vs direction	Paired	Patch
Primary	ACM2278	-0.000026	-0.000030	+0.000163	-0.000026
Primary	CDE1846	-0.000181	-0.001122	-0.001166	-0.000154
Primary	CMP2946	+0.000000	+0.000000	+0.000000	+0.000000
Primary	MBG3183	+0.000000	+0.000000	+0.000000	+0.000000
Primary	PLJ1771	+0.000000	+0.000000	+0.000000	+0.000000
Behavioral	ACM2278	+0.000337	-0.000722	+0.005161	+0.002383
Behavioral	CDE1846	+0.000008	-0.000418	-0.000450	+0.001105
Behavioral	CMP2946	+0.000266	-0.001415	-0.002502	+0.003050
Behavioral	MBG3183	-0.000002	+0.000464	+0.000464	-0.000002
Behavioral	PLJ1771	+0.003726	+0.001013	+0.001013	+0.001989

The dose table shows selected-minus-dictionary effects on incident-window days, its descriptive interval, available/planned positive pairs, and the paired window contrast. A dash denotes unavailable full-population estimation, not zero.

Family	Dose	Vs direction	95% interval	Pairs	Paired
Primary	0	+0.000000	[+0.000000, +0.000000]	73/73	+0.000000
Primary	0.25	–	–	72/73	–
Primary	0.5	-0.000076	[-0.000247, +0.000020]	73/73	-0.000064
Primary	0.75	-0.000187	[-0.000520, +0.000000]	73/73	-0.000227
Primary	1	-0.000230	[-0.000679, +0.000000]	73/73	-0.000201
Behavioral	0	+0.000000	[+0.000000, +0.000000]	73/73	+0.000000
Behavioral	0.25	-0.000410	[-0.000669, -0.000207]	73/73	-0.000372
Behavioral	0.5	-0.000716	[-0.001473, -0.000168]	73/73	-0.000713
Behavioral	0.75	-0.000420	[-0.000855, +0.000021]	73/73	-0.000257
Behavioral	1	-0.000216	[-0.000939, +0.000556]	73/73	+0.000737

The missing slot is ACM2278:239, primary dictionary control at strength 0.25, position 420. The same failure was reproduced from saved states and the fixed 32-candidate sequence. The reporting pilot matched all 95 arrays of a previously passing receiver and all 37 common arrays of the failure diagnosis exactly. All five full folds then passed their scoped independent checks. Separate aggregation reconstructs shifted-target means, pair weights, five-user means, coverage and bootstrap intervals from the raw arrays. The attribution-link audit checks the exact canonical deltas and selected activity on all 73 positive days. These checks establish implementation consistency; they do not supply new users, new datasets, or a replication under alternative donor policies.

A.2 Historical corrected r6.2 cache analysis

In the earlier cached pipeline, five corrected layer-18 dictionaries and their discovery-only feature rankings passed independent fitting, ranking and attribution checks. Each dictionary is transductive: unlabelled inputs from its held-out user enter fitting, while that user's examples are excluded from ranking. The primary five-feature sets place most discovery activation mass on profile positions, but activate very sparsely on the corresponding held-out users:

Held-out user	Days	Disc. profile (%)	Held-out entries	Held-out profile (%)
ACM2278	6	99.87	2	0.00
CDE1846	45	70.25	7	0.00
CMP2946	20	99.90	18	21.39
MBG3183	1	100.00	0	–
PLJ1771	1	99.97	1	0.00

"Entries" counts nonzero feature–token pairs, not distinct active tokens. Across 73 held-out incident-window days there are 28 such entries over 23,082 token positions: at most 0.122% of positions carry any selected activation. MBG3183 has zero mass, so its location fraction is undefined. Averaging the four defined users

gives 94.65% mass at behavior-related positions; this is the location of a small residual activation, not evidence of a broadly active behavior mechanism. In CDE1846 that mass occurs at session-count positions. Within each dictionary, held-out mean activation is at most 0.256% of its discovery mean. These post-hoc activity diagnostics indicate limited transfer of the selected family; they do not establish the cause or absence of other model features. The separately ranked behavioral family retains its separate scope.

This result describes the historical activation cache under corrected selection. A 64-example cache/live comparison failed the declared intervention screens. Controlled repeats are exact under fixed batch composition, but changing batch and padding changes hidden states and sparse supports. Cached attribution is therefore retained separately from the fresh canonical analysis in Section 7.3; the failed compatibility check is not relaxed. The fresh pipeline changes extraction and fitted dictionaries, so the difference from 28 historical entries to 20 fresh entries cannot be attributed to label correction alone.

B Relation to earlier methods

Anomaly detection supplies the one-class learning problem (Chandola et al., 2009); sequence and log models provide prior detector approaches (Du et al., 2017; Tuor et al., 2018; Guo et al., 2021). Shortcut learning and security-evaluation work motivate auditing the information behind apparent predictive success (Geirhos et al., 2020; Lapuschkin et al., 2019; Arp et al., 2022; Landauer et al., 2024). Sparse autoencoders provide candidate decompositions (Cunningham et al., 2023; Gao et al., 2024), while activation-patching methodology motivates matched interventions and careful interpretation (Zhang & Nanda, 2023; Heimersheim & Nanda, 2024). These established methods are the foundation of the audit. Applying them here does not itself establish a new sparse-coding algorithm, causal-identification theorem, or superior detector.

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